

Linear programming for T -avoiding spherical codes

Peter Boyvalenkov¹, Danila Cherkashin², and Peter Dragnev³

^{1,2}Institute of Mathematics and Informatics, Bulgarian Academy of Sciences, Sofia, Bulgaria*

³Department of mathematical sciences, Purdue University Fort Wayne, Fort Wayne, USA[†]

April 19, 2026

Abstract

This note summarizes recent results by the authors on T -avoiding spherical codes and designs, i.e. codes and designs whose inner products avoid a certain set $T \subset [-1, 1)$. We discuss the methods used to obtain these results and list open questions in this field.

Key Words : Spherical codes, Linear programming.

1 Introduction

Let $\mathbb{S}^{n-1} = \{x = (x_1, \dots, x_n) : x_1^2 + \dots + x_n^2 = 1\}$ be the unit sphere in n dimensions. A finite set $C \subset \mathbb{S}^{n-1}$ is a *spherical* (n, N, s) -code if $|C| = N$ and

$$s = s(C) = \max\{\langle x, y \rangle : x, y \in C, x \neq y\}.$$

A spherical (n, N, s) -code can be also referred to as an s -code. Denote by

$$I(C) := \{\langle x, y \rangle : x, y \in C, x \neq y\}$$

the set of all inner products of distinct points of C , so that $s = \max I(C)$.

Definition 1.1. Let $T \subset [-1, s)$. A spherical s -code $C \subset \mathbb{S}^{n-1}$ is called *T -avoiding* if $I(C) \cap T = \emptyset$.

The next classical definition is the following

*{peter, jiocb}@math.bas.bg

[†]dragnevp@purdue.edu

Definition 1.2. A spherical τ -design is a spherical code $C \subset \mathbb{S}^{n-1}$ such that

$$\int_{\mathbb{S}^{n-1}} p(x) d\sigma_n(x) = \frac{1}{|C|} \sum_{x \in C} p(x) \quad (1)$$

(σ_n is the normalized surface measure) holds for all polynomials $p(x) = p(x_1, x_2, \dots, x_n)$ of total degree at most τ .

A design C is called T -avoiding if $I(C) \cap T = \emptyset$.

Definition 1.3. Given a function $h : [-1, 1] \rightarrow (-\infty, +\infty]$, continuous on $(-1, 1)$, we define the h -energy of a spherical code C as

$$\mathcal{E}^h(C) := \sum_{x \neq y \in C} h(\langle x, y \rangle). \quad (2)$$

A code $C \subset \mathbb{S}^{n-1}$ is called T -avoiding universally optimal if for any absolutely monotone (i.e., all derivatives are positive) potentials h , it minimizes the h -energy among all T -avoiding codes of cardinality $|C|$.

Papers [BC25, BCD25, BD26, BCD⁺26] provide a few explicit examples of **largest T -avoiding codes**, **smallest T -avoiding designs** and **T -avoiding universally optimal codes**. These examples are usually contained in extremal even unimodular lattices. Here we summarize and highlight the most interesting cases and also comment on the way they are obtained.

2 Linear programming for maximal codes on the sphere

Let us introduce the classical linear programming (LP) techniques on the sphere. We call a function $f : [-1, 1] \rightarrow \mathbb{R}$ *positive definite* if

$$\sum_{x, y \in C} f(\langle x, y \rangle) \geq 0$$

for every code C . The idea is that the cone of positive definite functions has a nice basis, namely Gegenbauer polynomials $P_i^{(n)}(t)$, normalized by $P_i^{(n)}(1) = 1$. Such polynomials are orthogonal with the weight $w(t) = (1 - t^2)^{(n-3)/2}$ and satisfy the following three-term recurrence relation

$$P_i^{(n)}(t) = \frac{(n + 2i - 4)tP_{i-1}^{(n)}(t) - (i - 1)P_{i-2}^{(n)}(t)}{n + i - 3}. \quad (3)$$

Every function $f \in L^2([-1, 1])$ has the Gegenbauer expansion

$$f(t) = \sum_{i=0}^{\infty} f_i P_i^{(n)}(t), \quad (4)$$

where

$$f_i = \int_{-1}^1 f(t) P_i^{(n)}(t) \cdot w(t) dt \Big/ \int_{-1}^1 \left(P_i^{(n)}(t) \right)^2 \cdot w(t) dt .$$

The following theorem is the main tool of LP for T -avoiding codes.

Theorem 2.1 (Classical LP for codes [DGS77]). *Let $T \subset [-1, s)$ and $f(t)$ be a polynomial with real coefficients such that*

(A1) $f(t) \leq 0$ for all $t \in [-1, s] \setminus T$;

(A2) $f_0 > 0$ and $f_i \geq 0$ for all i in the Gegenbauer expansion (4).

Then

$$|C| \leq \frac{f(1)}{f_0}$$

for any T -avoiding spherical s -code $C \subset \mathbb{S}^{n-1}$.

For a given code C define its i -th moment $M_i(C)$ by

$$M_i(C) = \sum_{x, y \in C} P_i^{(n)}(\langle x, y \rangle).$$

By definition $M_i(C) \geq 0$ and $M_1(C) = M_2(C) = \dots = M_\tau(C) = 0$ if and only if C is a τ -design.

The corresponding main LP theorem for T -avoiding designs in the following

Theorem 2.2 (Classical LP for designs [DGS77]). *Let $\tau \in \mathbb{N}$, $T \subset [-1, 1)$ and $f(t)$ be a polynomial with real coefficients such that*

(B1) $f(t) \geq 0$ for all $t \in [-1, 1] \setminus T$;

(B2) $f_i \leq 0$ for all $i \geq \tau + 1$ in the Gegenbauer expansion (4).

Then

$$|C| \geq \frac{f(1)}{f_0}$$

for any spherical T -avoiding τ -design $C \subset \mathbb{S}^{n-1}$.

Analogously, LP can be applied to obtain bounds on h -energy as follows.

Theorem 2.3 (LP for energy [BD26]). *Let $h(t)$ be a potential function and $f(t)$ be a polynomial with real coefficients such that*

(E1) $f(t) \leq h(t)$ for every $t \in [-1, 1] \setminus T$;

(E2) $f_i \geq 0$ for all $i \geq 1$ in the Gegenbauer expansion (4).

Then

$$\mathcal{E}^h(C) \geq N^2 \left(f_0 - \frac{f(1)}{N} \right) \tag{5}$$

for every T -avoiding spherical code $C \subset \mathbb{S}^{n-1}$ of cardinality $|C| = N$.

If a code C attains equality in any of Theorems 2.1–2.3, then it often satisfies many additional properties, such as a prescribed inner product set $I(C)$, and sometimes even prescribed distance distribution or design properties.

3 Specifics in T -avoiding case

Here we list methods leading to a good polynomial $f(t)$ in Theorems 2.1, 2.2 and 2.3. Note that the polynomials arising in the most famous applications of these theorems — namely, the polynomials establishing the exact values of the kissing numbers (i.e., maximal $1/2$ -codes) in dimensions 8 and 24 (240 and 196 560, respectively) — can be obtained by any of the methods described in Subsections 3.1–3.3.

3.1 Reverse engineering

Starting with a good code C , one may construct suitable $f(t)$ by the following procedure. Let $I_i \subset I(C)$ and $I_b \subset I(C)$ be the sets of interior points and boundary points of $[-1, s] \setminus T$, respectively. The first attempt is to consider

$$f(t) = \prod_{r \in I_i} (t - r)^2 \prod_{r \in I_b} (t - r);$$

clearly, f or $-f$ satisfies the conditions of Theorem 2.1.

Theorem 3.1 (Theorem 6.1 in [BC25]). *Let $C \subset \mathcal{S}^{47}$ be a $(-1/3, -1/6) \cup (1/6, 1/3)$ -avoiding 11-design. Then $|C| \geq 52\,416\,000$. This bound is achieved by minimal vectors of an extremal even unimodular lattice in $\mathbb{R}^{48}$.*

Next step is to multiply such f by a polynomial of small degree, say $t + a$ or $t^2 + bt + c$ and then optimize over a , b , and c .

Theorem 3.2 (Theorem 4.2 in [BCD⁺26]). *Let $C \subset \mathbb{S}^{31}$ be a $(0, 1/4)$ -avoiding $(1/2)$ -code that is (at least) a spherical 3-design. Then $|C| \leq 146\,880$. This bound is achieved by minimal vectors of an extremal even unimodular lattice in $\mathbb{R}^{32}$.*

The vanishing of certain moments of the code C provides additional freedom in constructing $f(t)$. We have found an example where using additionally $M_7(C) = 0$ (with a putative optimal non-antipodal 5-design with $M_6(C) > 0$) allows to reduce T significantly.

Theorem 3.3 (Theorem 5.4 in [BCD25]). *The cardinality of an arbitrary $(-13/23, -7/23) \cup (5/23, 6/23)$ -avoiding 11/23-code in $\mathbb{R}^{23}$ is at most 48 600. This bound is achieved by a certain code inside Leech lattice.*

3.2 Brute-force method

The dual linear program to Theorem 2.1 is

$$\begin{aligned} \max c : \quad & \sum_{i=1}^k c_i P_i^{(n)}(1) \leq 1 - c, \\ & \sum_{i=1}^k c_i P_i^{(n)}(t) \leq -c, \quad \text{for } t \in [-1, s] \setminus T, \\ & c_i \geq 0, \quad \text{for } 1 \leq i \leq k. \end{aligned}$$

The corresponding upper bound on the cardinality of a code is $1/c_{\max}$.

Odlyzko and Sloane [OS79] used a brute-force approach with $k = 30$, and noted that for $n \leq 24$ the best polynomial is actually of degree at most 14. Technically, the approach is to replace the condition

$$\sum_{i=1}^k c_i P_i^{(n)}(t) \leq -c, \quad \text{for } t \in [-1, s] \setminus T$$

with K conditions

$$\sum_{i=1}^k c_i P_i^{(n)}(t_j) \leq -c, \quad \text{for } t \in [-1, s] \setminus T,$$

where $t_1, \dots, t_K$ is a subdivision of $[-1, s] \setminus T$ and then slightly repair (if necessary) the minimizing polynomial, taking into account subintervals $[t_i, t_{i+1}]$.

In practice, we use this approach to check that we have not missed a possibility from Subsection 3.1. For example, the corresponding program for Theorem 3.2 is as follows:

$$\begin{aligned} \max c : \quad & \sum_{i=1}^k c_i P_i^{(n)}(1) \leq 1 - c, \\ & \sum_{i=1}^k c_i P_i^{(n)}(t) \leq -c, \quad \text{for } t \in [-1, 0] \cup [1/4, 1/2], \\ & c_i \geq 0, \quad \text{for } 4 \leq i \leq k. \end{aligned}$$

The uniform choice of 500 nodes t_j and search for a degree-16 polynomial (corresponding to K and k in the program, respectively) gives

$$1/c_{\max} = 146880.0125 \dots,$$

which allows us to guess that there is a good polynomial.

On the other hand, if we only require C to be a 2-design ($c_i \geq 0$ for $3 \leq i \leq k$ in the program), then $1/c_{\max} = 159835.084 \dots$, and the described LP method does not work in this form. Since after such a simulation we switch to a rigorous proof we may not care about the accuracy of the calculations.

3.3 Levenshtein approach

Let $T = (a, b)$ for simplicity. Consider a polynomial

$$f(t) = (t - a)(t - b)(t - s)g(t),$$

where $g(t)$ is non-negative on $[-1, 1]$. Then f is a suitable polynomial for LP techniques for T -avoiding codes.

The following classical theorem was used by Levenshtein [Lev79] (see also [Lev83, Lev98]) in order to get a convenient representation of such polynomials g with bounded degree.

Theorem 3.4 (Lukács, see [Sze75]). *Let $g(t)$ be a polynomial of degree m , such that $g(t) \geq 0$ for $t \in [-1, 1]$. Then $g(t)$ can be represented in the form*

$$g(t) = \begin{cases} U^2(t)^2 + (1 - t^2)V^2(t), & \text{if } m \text{ is even,} \\ (1 + t)W^2(t) + (1 - t)Z^2(t), & \text{if } m \text{ is odd.} \end{cases} \quad (6)$$

Here $U(t)$, $V(t)$, $W(t)$, and $Z(t)$ are real polynomials such that the degrees of the single terms on the right-hand side do not exceed m .

For simplicity suppose that $m = 2k$ is even (odd case is a bit more technical). Put

$$U(t) = \sum_{i=0}^k u_i P_i^{(n)}(t), \quad V(t) = \sum_{i=0}^k v_i P_i^{(n)}(t).$$

Levenshtein's approach is to consider a polynomial g of degree $2k$ which gives a local extremum of

$$\frac{f(1)}{f_0} = F(u_0, u_1, \dots, u_k, v_0, v_1, \dots, v_k).$$

among all polynomials of degree at most $2k + 3$. The application of the Lagrange multiplier method immediately gives $v_0 = \dots = v_k = 0$ and

$$R\mathbf{u} = \mathbf{j}, \quad (7)$$

where $\mathbf{j}$ is the all-ones vector and R is a matrix defined by

$$R_{ij} = \left\langle (t - a)(t - b)(t - c)P_i^{(n)}(t)P_j^{(n)}(t), 1 \right\rangle.$$

The three-term recurrence relation (3) implies that R is a 7-diagonal matrix.

Summing up, we found the polynomial

$$f = (t - a)(t - b)(t - c) \left(\sum_{i=0}^k u_i P_i^{(n)}(t) \right)^2$$

that maximizes the quantity $f(1)/f_0$ over all polynomials of degree at most $2k + 3$. It is rather surprising that, for a large number of good codes C , the corresponding extremal polynomial f has nonnegative Gegenbauer coefficients f_i in (4).

In the following theorem we use this approach with $k = 2$.

Theorem 3.5 (Theorem 5.1 in [BCD25]). *Let*

$$T \in \{(-3/5, -1/3), (-1/3, -1/15), (-1/15, -1/5)\}.$$

The cardinality of a T -avoiding 7/15-code in $\mathbb{R}^{23}$ is at most 47104. This bound is achieved by a certain code inside Leech lattice.

3.4 Application to universal optimality

Once we have a good polynomial f from Subsections 3.1–3.3, we can try to generalize the bound to the corresponding universal optimality result.

For this purpose, we use Hermite interpolation of h at the roots of f (counted with their multiplicity) so as to guarantee $H_f(t) \leq h(t)$ for $t \in [-1, 1] \setminus T$, where H_f is the resulting Hermite interpolation polynomial. In order to prove the positive definiteness of the interpolation polynomial H_f we use a Newton-type interpolation formula

$$H_f(t) = h(t_1) + \sum_{r=1}^{\deg f} h[t_1, \dots, t_{r+1}] \prod_{j=1}^r (t - t_j),$$

where $\{t_j\}$ is the interpolation multiset and $h[t_1, \dots, t_{r+1}]$ are the corresponding divided differences. This breaks the problem into two independent parts. First, the divided differences $h[t_1, \dots, t_{r+1}]$ are nonnegative provided that h is absolutely monotone. Second, it suffices to prove that the partial products

$$\prod_{j=1}^r (t - t_j)$$

are positive definite, which allows us to apply Theorem 2.3. In practice, a good f often leads to the positive definiteness of the partial products since the last (and most difficult) partial product is the polynomial f itself (this phenomenon was already observed in [BDH⁺16]; see also [BCD25]).

Theorem 3.6 (Theorem 6.1 in [BCD⁺26]). *Let h be an absolutely monotone potential with $h^{(8)} > 0$ in $(-1, 1)$ and C be a T -avoiding code in $\mathbb{R}^{32}$ for $T = (-1/2, -1/4) \cup (1/4, 1/2)$. Then*

$$\begin{aligned} E_h(C) \geq 146\,880 & \left[h(-1) + 1\,240h\left(-\frac{1}{2}\right) + 31\,744h\left(-\frac{1}{4}\right) + 80\,910h(0) \right. \\ & \left. + 31\,744h\left(\frac{1}{4}\right) + 1\,240h\left(\frac{1}{2}\right) \right]. \end{aligned} \tag{8}$$

This bound is attained by the code formed from the minimal-norm vectors of any extremal even unimodular lattice in $\mathbb{R}^{32}$.

4 Open problems

Semidefinite programming (SDP) is a natural generalization of LP. An SDP method for spherical codes was developed by Bachoc and Vallentin in [BV08]. We are interested in analogous SDP techniques for the T -avoiding setting. To the best of our knowledge, the SDP approach has not yet been applied to spherical designs. One may hope that the T -avoiding setup is more amenable in the design case.

Shrinking T . Having a result for some set T , it is natural to ask for a minimal (with respect to inclusion) set $T' \subset T$ for which the same inequality remains valid. A first step is to investigate whether the endpoints of the intervals forming T can be moved inward.

Explicit formulas in the Levenshtein approach. In the case of s -codes, Levenshtein derived an explicit formula for the solution of (7), namely

$$u_i = \left(\binom{n+i-1}{i} - \binom{n+i-3}{i} \right) \begin{vmatrix} P_i^{(n)}(s) & P_{k+1}^{(n)}(s) \\ P_i^{(n)}(1) & P_{k+1}^{(n)}(1) \end{vmatrix}.$$

We are interested in an analogous formula for (a, b) -avoiding s -codes, which would be a first step toward theoretical asymptotic results in the T -avoiding setting.

Funding. The first author is supported, in part, by Bulgarian NSF grant KP-06-N72/6-2023. The second author is supported, in part, by Bulgarian NSF grant KP06-Austria/8-2025. The third author is supported, in part, by the Lilly Endowment.

References

- [BC25] P. Boyvalenkov and D. Cherkashin. The kissing number in 48 dimensions for codes with certain forbidden distances is 52 416 000. *Results in Mathematics*, 80, 2025. Article 3.
- [BCD25] P. Boyvalenkov, D. Cherkashin, and P. Dragnev. Universal optimality of T -avoiding spherical codes and designs. *arXiv preprint arXiv:2501.13906*, 2025.

- [BCD⁺26] P. Boyvalenkov, D. Cherkashin, P. Dragnev, D. Yorgov, and V. Yorgov. On t -avoiding spherical codes and designs in 32-dimensional Euclidean space. *Mathematics and Education in Mathematics*, 55(2):382–392, 2026.
- [BD26] P. G. Boyvalenkov and P. D. Dragnev. Energy of codes with forbidden distances in 48 dimensions. In *Recent Advances in Approximation and Potential Theory: In Honor of Edward Saff*, pages 157–172. Springer Nature Switzerland, 2026.
- [BDH⁺16] P. Boyvalenkov, P. Dragnev, D. Hardin, E. Saff, and M. Stoyanova. Universal lower bounds for potential energy of spherical codes. *Constructive Approximation*, 44:385–415, 2016.
- [BV08] C. Bachoc and F. Vallentin. New upper bounds for kissing numbers from semidefinite programming. *Journal of the American Mathematical Society*, 21(3):909–924, 2008.
- [DGS77] P. Delsarte, J.-M. Goethals, and J. J. Seidel. Spherical codes and designs. *Geometriae Dedicata*, 6:363–388, 1977.
- [Lev79] V. I. Levenshtein. On bounds for packings in n -dimensional Euclidean space. *Doklady Akademii Nauk*, 245(6):1299–1303, 1979.
- [Lev83] V. I. Levenshtein. Bounds for packings of metric spaces and some their applications. *Problemy Kiberneticki*, 40, 1983. In Russian.
- [Lev98] V. I. Levenshtein. Universal bounds for codes and designs. In V. S. Pless and W. C. Huffman, editors, *Handbook of Coding Theory*, chapter 6, pages 499–648. Elsevier, Amsterdam, 1998.
- [OS79] A. M. Odlyzko and N. J. A. Sloane. New bounds on the number of unit spheres that can touch a unit sphere in n dimensions. *Journal of Combinatorial Theory, Series A*, 26(2):210–214, 1979.
- [Sze75] G. Szego. *Orthogonal Polynomials*. AMS, Providence, RI, 1975.